

Project Proposal

Multi-Region Brain Tumor Segmentation Using the BraTS 2024 Adult Glioma Dataset

a. Student Information

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b. Chosen Topic

This project is based on the Brain Tumor Segmentation (BraTS) 2024 challenge. The 2024 edition focuses on multi-region segmentation of adult diffuse gliomas from multi-parametric MRI (T1w, T1ce, T2w, T2-FLAIR), covering both pre-treatment and post-treatment imaging scenarios.

c. Problem Statement and Motivation

Accurate MRI segmentation of adult gliomas is essential for treatment planning, but manual segmentation is slow and highly variable. Automating this task is challenged by high tumor heterogeneity across MRI modalities, severe class imbalance, and varying institutional acquisition protocols.

d. Dataset

The dataset that will be used is BraTS 2024 Adult Glioma (post-treatment task)

e. Proposed Method and Existing Implementations

The segmentation model will be implemented in PyTorch, with MONAI used for dataloaders and transforms. Two architectural options are under consideration. The first is a 3D U-Net processing full volumetric patches, which captures spatial context across all three axes simultaneously, the standard approach for BraTS. The second is a 2D U-Net operating on individual axial slices, which is faster and less memory-intensive but loses inter-slice spatial context. Both options will be trained using Dice Focal Loss, AdamW, and a cosine annealing schedule, and the proposed Monte Carlo Dropout innovation applies identically to either architecture.

f. Evaluation Methodology

Model performance will be assessed using the standard BraTS 2024 evaluation metrics: Dice and 95th Percentile Hausdorff Distance.

g. Proposed Innovation

The main innovation of this project is uncertainty-aware segmentation via Monte Carlo Dropout. Standard segmentation models produce a single hard prediction per voxel with no indication of model confidence, which is a meaningful limitation in clinical practice. We propose to keep dropout active at inference time and perform multiple stochastic forward passes, aggregating the outputs to produce both a final segmentation and a per-voxel uncertainty map based on predictive entropy across the samples. The modification requires only the addition of dropout layers within the encoder and decoder blocks. We will validate the uncertainty maps by examining whether high-uncertainty regions correspond to known sources of segmentation difficulty, such as tumor boundaries and heterogeneous contrast enhancement zones.